



DEPARTMENT OF TECHNICAL EDUCATION

A

PROJECT SYNOPSIS REPORT ON

“LIBRARY MANAGE SYSTEM PROJECT “

DIPLOMA IN COMPUTER SCIENCE AND ENGINEERING

PROJECT ASSOCIATES

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This is certifying that **MOHAMMED MUNANZAR(180CS24018)** Students of department of computer science & engineering have successfully completed **"LIBRARY MANAGE SYSTEM PROJECT "** and submitted the synopsis in partial fulfilment of the requirement prescribed by board of technical examination, department of technical education, Bangalore for the academic year 2026-2027.

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INTRODUCTION

The Library Management System is a computer-based software application designed to manage and automate the various activities performed in a library. A library contains a large collection of books along with information about students, teachers, members, issued books, returned books, and fines.

Managing all these records manually using registers and paperwork can be difficult, especially when the number of books and members is large.

The proposed system provides a centralized platform where all library-related information can be stored and managed systematically. The librarian can add new books, update book information, register members, issue and return books, check book availability, and maintain fine records. The system helps reduce manual work and provides quick access to required information.

The main purpose of this project is to make library management simple, fast, accurate, and organized. It also helps maintain records securely and reduces the chances of losing or incorrectly entering important information.

PROBLEM STATEMENT

In a traditional library, most activities are performed manually using registers, files, and paper records. Maintaining a large number of book and member records manually requires considerable time and effort. When a student wants to find a particular book, the librarian may have to search through registers or shelves to check its availability.

Manual management can also create problems such as duplicate records, incorrect entries, missing information, difficulty in tracking issued books, and errors while calculating fines. Keeping track of return dates can also be difficult when many books are issued to different members.

Therefore, there is a need for a computerized system that can manage library information efficiently. The Library Management System solves these problems by storing all important information in a structured database and providing different modules for managing library operations.

OBJECTIVES

The main objective of the Library Management System is to automate the major activities of a library and reduce dependence on manual record keeping. The system is designed to provide an easy and efficient method for managing books and members.

The important objectives are to maintain complete book records, store member information, manage book issuing and returning, check book availability, and calculate fines for late returns. The system also aims to reduce paperwork and minimize human errors.

Another objective is to provide quick searching and retrieval of information. Instead of searching through multiple registers, the librarian can enter a book name, author, category, or ID and obtain the required information quickly. The system also helps maintain accurate and updated records of all library transactions.

BOOK MANAGEMENT

The Book Management Module is one of the main components of the system. It is used to maintain complete information about all books available in the library. Each book can be assigned a unique identification number so that it can be easily searched and tracked.

The librarian can enter details such as Book ID, Book Name, Author Name, Publisher, Category, Edition, Price, Quantity, and Availability Status. If new books are purchased, their information can be added to the database. Existing information can also be updated whenever necessary.

The system can also help identify whether a book is currently available or has been issued to another member. This makes book management easier and helps the librarian maintain an accurate collection of library resources.

STUDENT MEMBER MANAGEMENT

The Student or Member Management Module is used to maintain information about people who are registered to use the library. Every member can have a unique membership or student ID.

The system can store information such as Student ID, Name, Course, Semester, Department, Contact Number, Email, and Membership Details. The librarian can register new members and update their information when required.

This module makes it easier to identify which member has borrowed a particular book. It also helps maintain a proper record of library users and their transactions. Having member information in a centralized database reduces the need for maintaining separate paper files.

BOOK ISSUE AND RETURN

The Book Issue Module manages the process of issuing books to students or other registered members. Before issuing a book, the system checks whether the book is available in the library.

When a book is issued, the system records important information such as Book ID, Member ID, Issue Date, Due Date, and Issue Status. The availability status of the book is then updated so that other users can know that the book is currently unavailable.

This module helps the librarian track which member has borrowed a particular book and when it is expected to be returned. It also reduces mistakes that can occur when issue details are recorded manually.

FINE MANAGEMENT

The Book Return Module is responsible for recording the return of books issued to members. When a member returns a book, the librarian can search for the corresponding issue record using the book ID or member ID.

The system records the return date and checks whether the book was returned on time. After the return is completed, the book's availability status is changed to available. If the book is returned late, the system can forward the information to the Fine Management Module.

This module provides an accurate record of returned books and makes it easier for the librarian to identify books that are still pending for return.

SEARCH AND AVAILABILITY

The Fine Management Module is used to calculate and maintain fines for books that are returned after their due date. Late returns can be identified by comparing the due date with the actual return date.

The system can calculate the number of delayed days and apply the fine amount according to the rules defined by the library.

The fine details can be stored along with the member's transaction record.

This reduces the need for manual calculations and helps avoid errors. It also provides a clear record of fines associated with different members and books.

CONCLUSION

The Library Management System provides a systematic and computerized approach to managing the activities of a library. It replaces many manual processes with an organized digital system that can store, process, and retrieve information efficiently.

The system manages important operations such as book management, member registration, book issuing, book returning, fine calculation, searching, and report generation. By maintaining these records digitally, the system helps reduce paperwork, save time, improve accuracy, and simplify the work of the librarian.

Overall, the project provides a practical solution for modern library management. It can be used as a foundation for developing a more advanced system with online reservations, notifications, barcode scanning, mobile access, and other digital library features.

